

Part03 - Bonus Questions

Question 4: Self-study report

Over the past few weeks, I've been exploring the basics of C# with a special focus on arrays, loops, exception handling, and enums.

I practiced writing clean code using for, foreach, and while loops, and learned how input validation helps avoid runtime errors.

I also discovered the difference between shallow and deep copies of arrays, and used methods like Sort, IndexOf, and Clear.

Through this journey, I've built a solid foundation that will help me dive into more advanced C# concepts like OOP, LINQ, and async programming.

Question 5: What's the default size of Stack and Heap?

What are the considerations?

Answer:

- In **.NET**, the default size of the **stack** is usually:
 - **1MB** per thread (on most systems).
- The **heap** size is much larger and managed dynamically by the **garbage collector**.
- **Considerations:**
 - Use the **stack** for small, short-lived variables (like local variables).
 - Use the **heap** for large or complex objects (e.g., arrays, class instances).
 - Stack is faster but limited in size, while heap is more flexible.

Question 6: What is Time Complexity?

Time Complexity refers to **how much time an algorithm takes to run** as a function of the input size n .

Complexity	Example Algorithm	Description
$O(1)$	Accessing array element	Constant time
$O(n)$	Loop through array	Linear time
$O(n^2)$	Nested loops	Quadratic time
$O(\log n)$	Binary search	Logarithmic time